



TECHNICAL UNIVERSITY OF MUNICH

SCHOOL OF COMPUTATION, INFORMATION AND
TECHNOLOGY
INFORMATICS

Master's Thesis in Informatics

**DRAFT - Thesis Proposal:
Something With Usability in PIX:E**

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DRAFT - Thesis Proposal:
Something With Usability in PIX:E
Thesis Proposal: Irgendetwas mit Usability
in PIX:E

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Disclaimer

I confirm that this Master's thesis is my own work and I have documented all sources and material used.

(Date)

(Rai Trouvain)

Acknowledgements

The acknowledgements

Abstract

English Abstract

Zusammenfassung

German Abstract

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1 Introduction

2 Background

3 Path-Based Diagrams

- basic idea already present in PaceMaker: select path in statechart and visualize data along path
- main design guideline: build upon high configurability already present in the system
- following sections describe functionalities, implementation choices and workflow/usage of these features implemented in pixe

3.1 Path Selection

3.1.1 Functionality

- selection of nodes in statechart triggers calculation of path connecting them
- selected path is highlighted (visual feedback)
- path is subsequently used for diagram generation

3.1.2 Implementation

- select start + end node, optionally intermediate -> calculate path using dijkstra
- `pathsApi.ts`: to allow for different algorithms in the future (e.g. specific pathfinding for nestedness and concurrency)

3.1.3 Workflow

- selection of at least two nodes
 - Ctrl + click
- path is calculated along nodes **in order of selection**
- if path is found, the corresponding nodes are highlighted
- path can be de-selected by removing node selection (clicking anywhere)

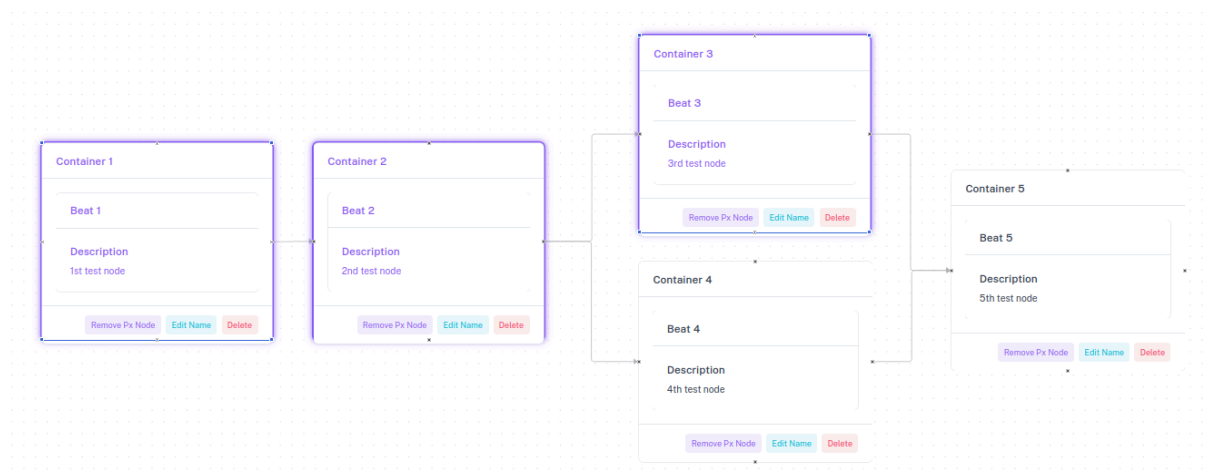


Figure 1: Path highlighted in state chart

3.2 Diagram Generation

3.2.1 Functionality

- expandable element on statechart page
 - usability: users can hide diagrams while working on statechart
- can create multiple diagrams and remove diagrams (red because diagrams do not persist)

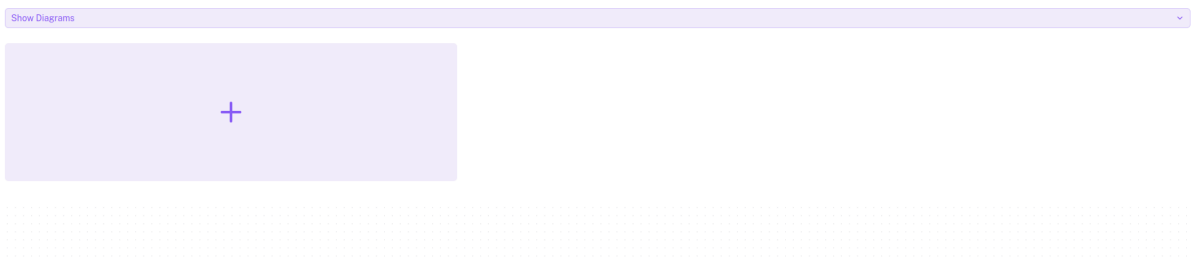


Figure 2: Upper section of the charts page with expandable element and UI for diagram creation

- selection of one or multiple component definitions for the y axis
- optionally selection of one component for the x axis (e.g. estimated playtime)
 - values are summed up along selected path
 - default: equal spacing

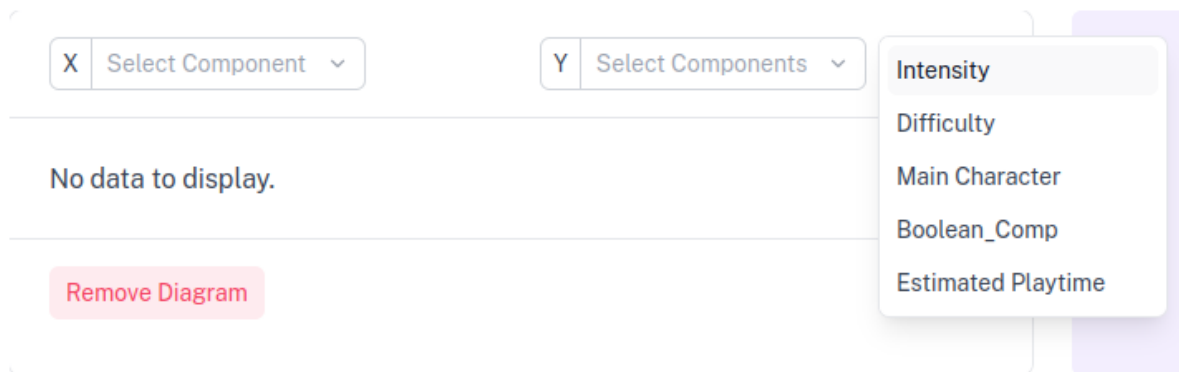


Figure 3: Selectors for components to be mapped to X/Y axes

- selected path along x axis
 - if no path selected: all nodes,  sorted



Figure 4: Diagram visualizing component along selected path

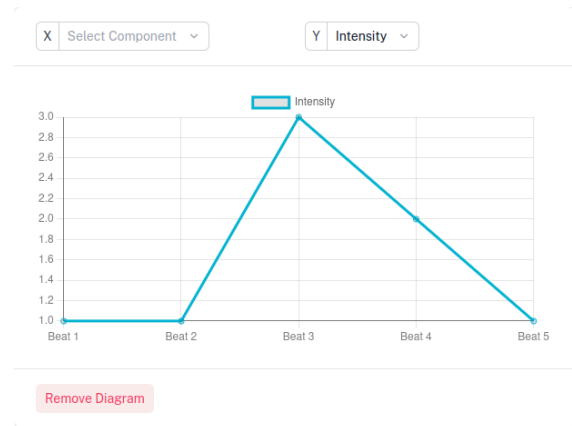


Figure 5: Diagram visualizing component across all nodes

3.2.2 Implementation

- [chartJS](#) library
- modular architecture: `PxDiagrams` as wrapper element
 - manages diagrams
 - currently: passes path (reactively), deletes diagrams
 - future work: path snapshots, different data types
- data is extracted from nodes once per diagram, switching between axis configurations is then done via different parsing configurations

3.2.3 Workflow

1. add new diagram
2. in any order:
 - select one or no component for x axis
 - select one or more component for y axis
 - select path
- changes in any of the configurations will be reflected live in the diagram

3.2.4 Limitations

- diagrams do not persist (reset when reloading page)
- only line diagrams for now
- same path selection for all diagrams

4 Locks and Keys

requirements/context:

- multiple keys possible per node
- multiple locks possible per node
- pathfinding and reachability analysis

4.1 Definition

(Ashmore and Nitsche 2007): “The puzzle is finding out what is an obstacle, what and where is a key to overcome it, and finally using the key to master the challenge.”

- in statechart: conditional transition

(Ashmore and Nitsche 2007): “Obstacles may not be passed until the player obtains some token (such as an item or skill)”

- keys as abstract tokens, different kinds to distinguish: keys/locks should have some category to specify this

(Dormans 2017): “Some locks are barriers that might be navigated without a key, but this crossing the barrier might be uncertain or impose a certain risk.”

(Ashmore and Nitsche 2007) present lock/key puzzles as a type of quest

- future work: model/analyze other types of quests (explicit, specific tasks, mission objectives)

4.2 Taxonomy

(Dormans 2017): “When you unlock a door, that door might remain unlocked forever (permanent), for a short period of time (temporary), or until it is relocked (reversible). Sometimes, a lock collapses after use, allowing the player only to pass once.”

-> locks: **permanent, temporary, reversible, collapsible**

(Dormans 2017): “Single-purpose keys can only be used to open a lock, and for nothing else, while multipurpose keys can also be used in different ways.”

-> keys: **single-purpose or multi-purpose**

- attribute type on keys: ability, item

(Dormans 2017): “Particular keys are the only thing that unlocks a particular lock, whereas several nonparticular keys might unlock a single lock.”

-> keys: **particular or non-particular**

- probably easier to understand when this is a property of a lock, i.e. key requirement

(Dormans 2017): “Keys that are destroyed somehow in the process of unlocking a door are consumable, while keys that are not are persistent.”

-> keys: **consumable or persistent**

(Dormans 2017): “Levers and switches are the best example of keys that are fixed in place (and typically single purpose and particular as well).”

-> keys: **fixed or not**

4.2.1 Not Modeled Currently

(Dormans 2017): “Certain locks allow you to cross only in one direction (valves), while others can only be opened from one direction but traversed in two directions after they are opened (asymmetrical). [...] Valves do not always require a key.”

-> locks: **valve, asymmetrical**

- locks are assigned to edges, so valves can already be modeled. only question: what if one unlock opens it for both directions?

(Dormans 2017): “A safe lock is guaranteed to have a solution, while an unsafe lock is not.”

4.2.2 Data Model Overview

 **add diagram of data model**

4.3 Creating and Visualizing Locks and Keys

(Brown 2026): high-level visual representation. only focuses on locks and keys and where they are in relation to each other

to integrate into player experience chart: assign keys to nodes and locks to edges (to maintain clear structure)

4.3.1 Modeling Keys

- similar to PxComponents: definitions separate from instantiations in nodes
- multiple instances of same definitions possible per node

 **add screenshots of key definitions and assignments in UI + explain options**

4.3.2 Modeling Locks

- same split between definitions and instantiations
- instantiations on edges: additional UI/modal

 **add screenshots of lock definitions and assignments in UI + explain options**

4.4 Pathfinding with Locks and Keys

4.4.1 Basic Dijkstra with Locks and Keys

- goal: all locks assigned to edges in the path have to be unlocked by key(s) collected up to the edge in question
- inventory: track keys collected along path to determine whether encountered edges can be unlocked
 - inventory similar to (Aversa, Sardina, and Vassos 2015)
 - modeled via `PxKeySet` -> formally: mapping of key defs to counts
- basic form: inventory associated with a node reflects keys collected along shortest path to this node and keys made available in this node -> later extended (consumable keys)

4.4.1.1 Unlocking Function

- `canUnlock()`: given (multi-)sets of locks and keys, determine whether locks can be unlocked
 - added complexity due to an edge potentially having multiple locks, and each locks potentially being unlocked by multiple keys
 - exponential blowup, but no optimization so far as average case is expected to be less complex  citation

given set of keys in inventory K and locks on edge L :

```
canUnlock(K, L):
  if L = ∅
    return TRUE
  unlockingKeysets = required(l1) × ... × required(ln)
  if ∃ U ∈ unlockingKeysets where ∀ u ∈ U. u ∈ K:
    return TRUE
  else
    return FALSE
```

4.4.1.2 Example

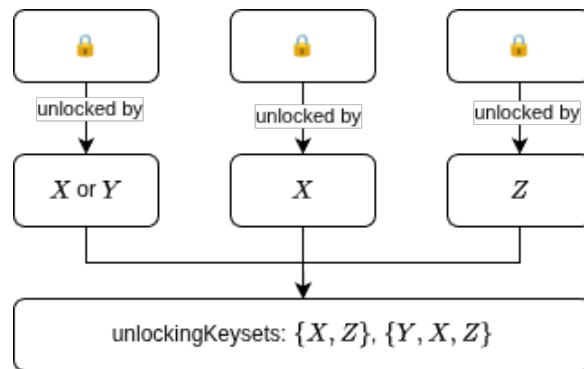


Figure 6: Calculation of unlocking keysets

- $I = \{X, Y\} \rightarrow \text{false}$
- $I = X, Y, Z \rightarrow \text{true}$

4.4.1.3 Full Algorithm

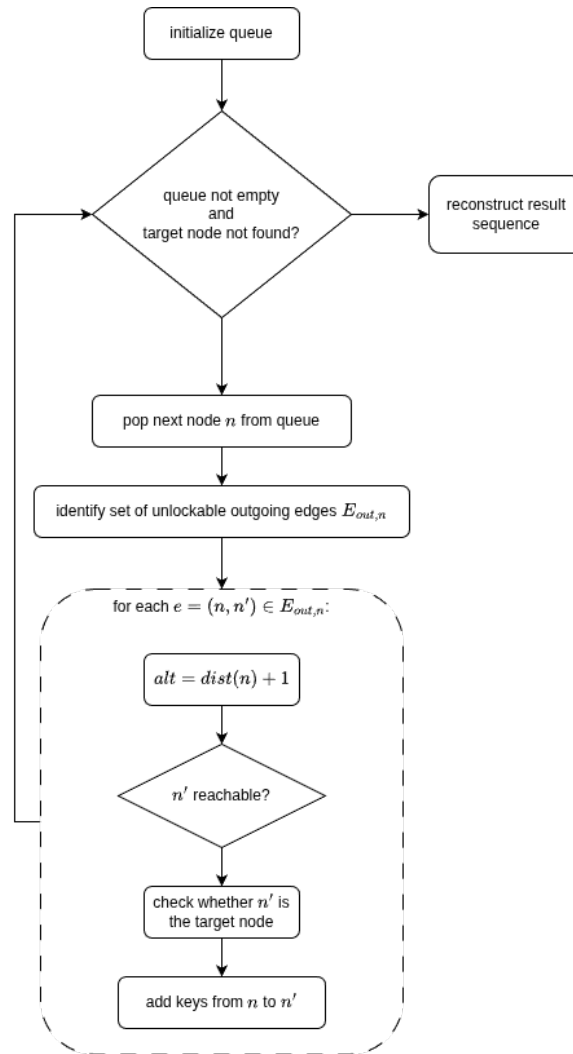


Figure 7: Calculation of unlocking keysets for key consumption

4.4.1.4 Example Iteration

-  add small example of iteration

4.4.2 Fixed Keys

- goal: fixed keys can only unlock locks on outgoing edges of the node they are assigned to

implementation:

- do not add keys to inventory of neighbors
-  add example

4.4.3 Soft Gates

- goal: soft gates **can** be passed without the corresponding key, but this should be indicated in the path highlighting

implementation:

- during pathfinding, ignore soft-gate locks
- after path has been found, re-trace it to determine whether keys are available or not
-  add example from pixe

4.4.4 Consumable Keys and Soft-Lock Detection

- goal: inventory should reflect usage of single-use keys
- variants of inventory -> idea of ‘parallel universes’
 - nodes exist in different realities if they have different inventories
 - assign each node a set of possible keysets

implementation:

- maintain set of `PxKeySets` per node
- in `canUnlock`, consider multiplicity of consumable keys
- for updating inventory of neighbor nodes: remove consumed keys from inventory
 - if there are multiple possible constellations, add multiple keysets

4.4.4.1 Pathfinding with Consumable Keys

Extending Unlocking Check

given multiset of keys in inventory K and locks on edge L :

```
canUnlock(K, L):
  if L = ∅
    return TRUE
  unlockingKeysets = required(l1) × ... × required(ln)
  if ∃ U ∈ unlockingKeysets where U ⊆ K:
    return TRUE
  else
    return FALSE
```

- note: subset of multiset

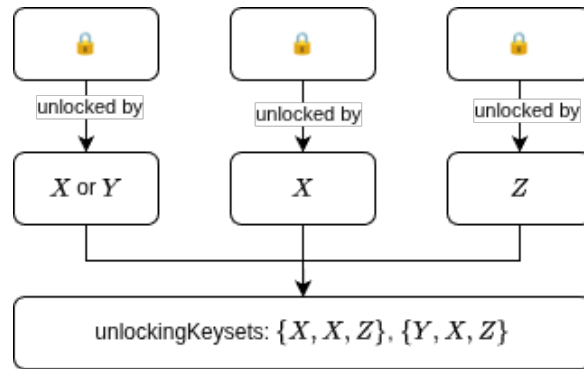


Figure 8: Calculation of unlocking keysets

- now: $I = \{X, Z\} \rightarrow \text{false}$

Removing Consumable Keys from Inventory

given inventory I containing multisets of keys K and locks on edge L :

```

canUnlock(I, L):
  if L = ∅
    return TRUE
  consumableRequirements = locks with at least one consumable key unlocking
  unlockingKeysets = required(l1) × ... × required(ln)
  updatedInventory = {}
  for K_i ∈ I:
    if ∃ U ∈ unlockingKeysets with K_i ⊆ U:
      updatedInventory.push(K_i - {key ∈ U. consumable(key)})
  return updatedInventory
  
```

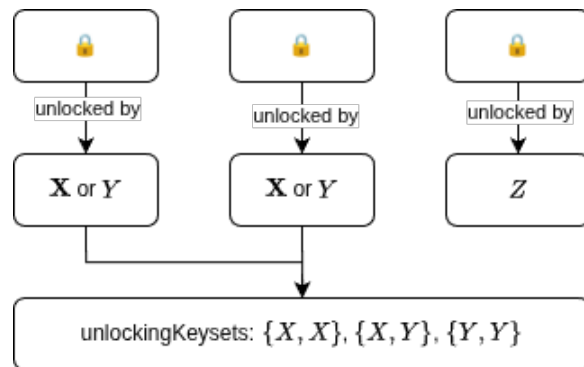


Figure 9: Calculation of unlocking keysets for key consumption

- $I = \{X, X, Z\} \rightarrow \{Z\}$

-  example in pixe for consumable keys

4.4.4.2 Soft-Locks

definition: “it is possible to reach the goal state from every reachable state” ([Mawhorter and Smith 2021](#)):

- here: not nodes, but *states* in which nodes are visited

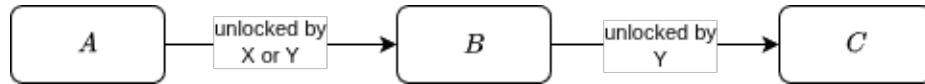


Figure 10: Calculation of unlocking keysets for key consumption

implementation:

- potential soft-lock state: node unlockable with some, but not all keysets
- when removing consumed keys: remove empty keysets entirely. then, compare inventory length before and after consumption
-  example in pixe for consumable keys

4.5 Bidirectional Edges and Backtracking

- goal: allow for longer/cyclic paths needed to collect keys
- another idea of parallel universes
-  add citation from julians paper?

implementation:

- alternative used here: nodes are re-added to queue iff they have a different inventory assigned to them than in previous encounters -> only duplicate nodes as needed
 - for each node, maintain inventories it has been visited with
 - only re-visit with a different inventory
 - add limiting constant for revisits
 - also maintain unlocked edges



Figure 11: Calculation of unlocking keysets for key consumption

- critical path: $A - B - A - C$

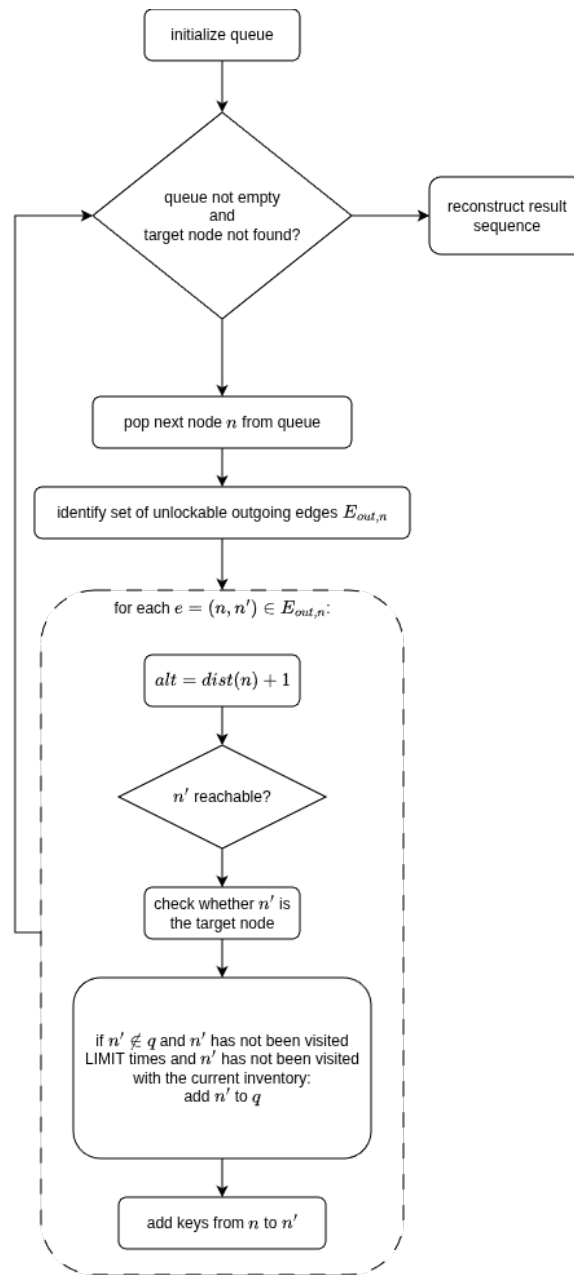


Figure 12: Calculation of unlocking keysets for key consumption

4.5.1 Example

-  example screenshot from pixe

4.5.2 Full Algorithm

-  make diagram of full algorithm?

4.6 Options/Settings

-  describe pathfinding options

4.7 Code Architecture

-  add diagram for code architecture?

4.8 Testing

-  describe

4.9 Limitations and Future Workflow

- difficulty estimation  add citation
- recommendations for lock/key types ([Dormans 2017](#))
- LLM-based consistency check with descriptions

5 Evaluation

- user study

5.1 User Study Design

5.1.1 Overall Setup

- task-based within-subject: StatePx vs Miro
 - two task variants -> 4 groups
- use case: zelda dungeons ([Summerville et al. 2016](#)) -> ensure realistic tasks. data pool big enough to choose 2 comparable dungeons per task. same game: consistency in terms of details like item/key types, dungeon structure, etc.
- target demographic: people with some understanding of game design -> recruitment: university lecture in games engineering study course, also computer science students who are familiar with videogames

5.1.2 Modeling Zelda Dungeons

5.1.2.1 General Modeling

- rooms numbered based on location in map (approx from bottom left to top right)

 [image showing room numbering](#)

5.1.2.2 pix:e vs Miro

pix:e: rooms: node. use in-built functionalities for keys/items and other room data (components). visualization: as described previously

Miro: rooms: sticky notes. write all information into sticky notes. heavier use of emoji encoding: to more easily differentiate between different key types/lock types

comparison: pix:e has more dedicated functionality + logic, but ui is less mature (zoomed out view, editing features). emoji encoding in Miro.

5.1.3 Solvability Tasks

- larger Zelda dungeons
- Link's Awakening LA_3, LA_4 ([Summerville et al. 2016](#))
- selection criteria: size (35 vs 41 nodes), number of keys/locks. comparable size + complexity (outgoing ranks of nodes)  [add details](#)

 [images: original maps/graphs, versions in Miro/pix:e](#)

- slight adaptations:  [add adaptations](#)
 - structural:  [add structural adaptations](#)

- information: remove puzzles, “vision” edges, enemies
- task: find as many issues -> specified kinds of issues

5.1.4 Pacing Analysis Tasks

- smaller Zelda dungeons
- Link’s Awakening LA_1, LA_2 ([Summerville et al. 2016](#))
- selection criteria: size, number of keys/locks. comparable size + complexity  add details
-  images: original maps/graphs, versions in Miro/pix:e
- adaptations:  add adaptations
 - structural:  add structural adaptations
 - information: remove puzzles, “vision” edges
 - additional information: completion time based on YouTube walkthrough (100% completion to get data for all rooms)  add ref to walkthroughs
- task: answer questions about pacing. includes comparisons of path sections, overall enemy distribution in nodes

5.1.5 Collected Data

- demographics: age, gender, background, occupation experience
-  add complete demographics
- recorded during tasks: issues found/questions answered + timestamps, comments, clarifications/hints, observations about tool use
- additionally: UEQ-S ([Schrepp, Hinderks, and Thomaschewski 2017](#)), questionnaire comparing Miro and StatePx

5.2 User Study Results

5.2.1 Data Processing

-  potentially more of a methodology section? check wiki
- data recorded in semi-structured or format -> parsing script to transform
- different sets of data:
 - demographics
 - task meta data per participant and task: total counts/points - based on protocol, may contain some human error
 - time series per participant and task: timestamp + individual answers, comments, etc - based on protocol, may also contain some human error

- UEQ-S
- comparison questionnaire

5.2.2 Data Analysis

5.2.2.1 Participants

- total number: 24
- demographics
- group breakdown

5.2.2.2 Results Solvability

- total number of identified issues
- correctly identified issues (f1)
- times to first identified issues

5.2.2.3 Results Pacing

- total number of answered questions
- correctly answered questions (f1)
- wrongly answered questions
- time per question

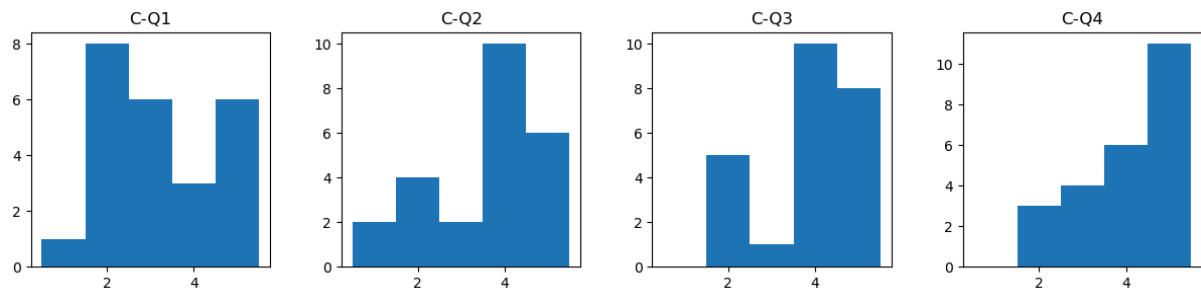
5.2.2.4 Results UEQ-S

- using provided analysis tool
- pragmatic quality (“relate to the tasks or goals the user aims to reach”): 0.823, hedonic quality (“related to pleasure or fun while using the product”): 1.385, overall: 1.104
-  contextualize
- weakest item: confusing vs clear
 - potentially relates to: UI quirks, lock visualization

5.2.2.5 Results Comparative Questions

- C-Q1: “Compared to Miro, the system made it easier to analyze dungeon designs. “
- C-Q2: “Compared to Miro, I felt more confident in my answers. “
- C-Q3: “Compared to Miro, I was able to complete the tasks more efficiently. “
- C-Q4: “If I were analyzing game levels in practice, I would prefer using the system over a general-purpose whiteboard.”

5-point Likert scale, fully disagree (1) to fully agree (5)



- overall, participants indicated preference for statepx over miro, mixed answers with a slight preference for pixie in first three questions
- in line with high hedonic quality reported in UEQ-S
- previous whiteboard experience correlated with lower preference for pix:e  [back](#)
this up with calculation

5.2.2.6 Validity of A/B Test

- compare results across groups -> statistical test
- compare demographics across groups

5.3 Limitations

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